A matched test of dye selection

Can one reference movie improve the choice of two spontaneously blinking dyes for a fixed-cell H2B-HaloTag experiment? Blink Atlas uses measured on-times, off-times and photon rates to estimate how many isolated signals a camera can record.

First request / An existing matched dataset

Could you share raw movies or SMAP localization tables for fixed cells from the same batch and microscope, imaged with at least two of JF630b, JF635b and JF639b? Two ligands permit a pairwise check. All three permit the two-choice comparison. Include preparation day, well, field, region area, labeling protocol, laser power, exposure, frame interval and pixel size. For tables, include frame, x/y, detected photons and lateral localization uncertainty with units.

If new imaging is needed

54 movies = 3 dyes × 2 labeling densities × 3 independent preparation days × 3 fields per condition.

Use fixed COS-7 cells expressing H2B-HaloTag. Split each preparation into matched wells, set sparse and dense labeling in advance, and randomize acquisition order. Keep 647 nm excitation, 25 ms exposure, frame interval, camera settings and scored acquisition time constant across the three ligands.

Freeze the choices before scoring

Within each day and density, reserve field 1 of each dye for the pilot. The model sees only the first 200 frames of reference JF630b. Both strategies receive this same pilot budget and two candidate slots. Rank only JF630b, JF635b and JF639b. Lock the pair before revealing fields 2 and 3. The comparison pair is always **JF630b + JF635b**. Use that fixed pair when the model abstains and record every abstention.

Endpoint and replication

Count localizations per μm² with lateral uncertainty at or below **20 nm**, using the same acquisition time. For per-axis estimates, require both axes to pass. Within each day and density, average each dye over its two scoring fields and take the higher yield of the two selected dyes. Report both strategies and their paired difference for each preparation day. Repeat on a new target or instrument once gains recur across independent preparations.

Measured photophysics from Holland et al., *Nature Methods* (2026), Extended Data Fig. 4. doi:10.1038/s41592-026-03062-5